

Aditya Dutta

Robotics | Embedded Systems
Rapid Prototyping | Leadership

Kolkata & Bengaluru, India

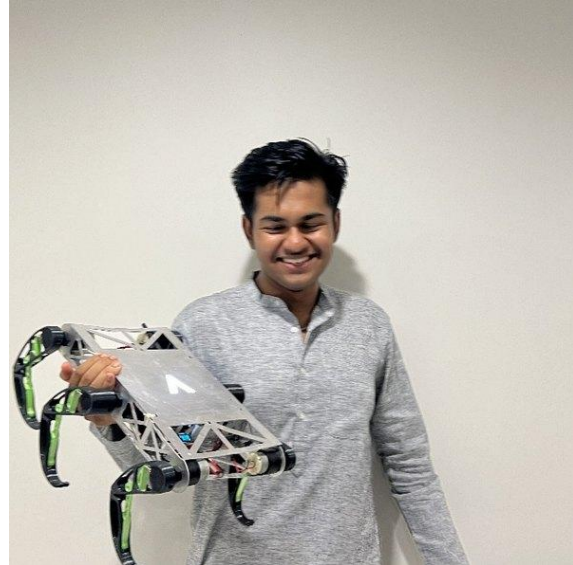
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Skills

Embedded & Electronics Systems Embedded C/C++, STM32 and ESP32 architectures, low-power firmware, peripheral interfacing, sensor integration, real-time I/O, hardware debugging.

PCB Design & Hardware Engineering Multi-layer PCB design (Altium), power systems (Li-ion charging and regulation), high-density integration, manufacturable layouts, signal integrity.

Robotics & System Integration End-to-end systems spanning compute, control and actuation; distributed architectures (Jetson + MCU); BLDC and servo motor control; feedback systems; sensor fusion.

Mechanical Design & Rapid Prototyping CAD modelling, mechanism design both bio-inspired and functional, 3D printing, laser-cut fabrication, iterative prototyping under constraint.

Software & Autonomy Stack Python, robotics software structuring, modular architectures, vision-based perception pipelines, hardware abstraction layers.

Control Systems & Mechatronics Closed-loop control, encoder feedback, motion control, system tuning, and handling real-world constraints of latency, noise and power.

Creative Engineering & Prototyping Zero-to-one system design, constraint-driven innovation, rapid iteration, translating abstract concepts into deployable hardware.

Leadership & Technical Direction System architecture planning, design reviews, mentoring university engineering teams, decomposing complex systems into executable subsystems.

Information

Undergraduate, Computer Science & Engineering

Manipal Institute of Technology, Bengaluru (MIT-B)

Third year · expected graduation 2028

Academic interests Robotics, embedded systems, autonomous platforms, product design and manufacturing.

Current roles President, RADAR Student Robotics Club.
Co-Founder & Technical Lead, Sylvara Inc.

Hometown Kolkata, West Bengal

Tools & Software

Autodesk Fusion 360 · Altium Designer · ROS 2 Jazzy · NVIDIA Isaac Sim · Jetson compute · MATLAB · Oracle SQL · Adobe Photoshop · BambuStudio & Cura · Linux terminal

Multi-layer PCB design · Advanced CAD · 3D printing · CNC & laser manufacturing · Manual PCBA · Rapid prototyping · Product conceptualisation

ONGOING RESEARCH

CatBot

BIOMIMETIC QUADRUPEL · DISTRIBUTED EDGE NODE

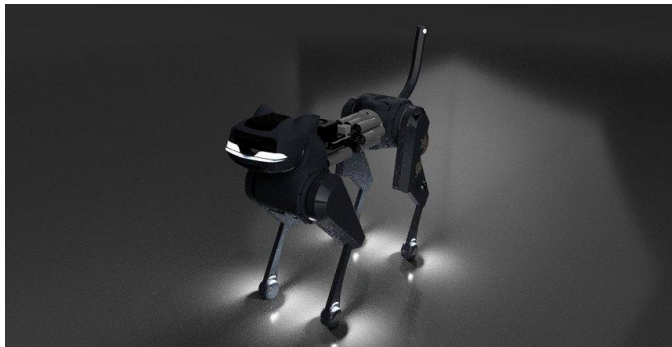
OBJECTIVE

Build a quadruped held to the physical envelope of the animal it copies — the footprint and all-up mass of a wildcat, under 5 kg — and give it enough onboard autonomy to work unsupervised where a wheeled platform cannot go.

SYSTEM OVERVIEW

CatBot is a 19-degree-of-freedom, 4.8 kg biomimetic quadruped built around a decentralised electronic nervous system. The obvious way to build a quadruped is one controller wired to every motor; that stops working the moment mass, latency and wiring matter. CatBot inverts it — every actuator carries its own FlexNode closing a tens-of-kilohertz current loop millimetres from the windings, and what runs down the body is a CAN-FD pair and power.

Above that sit two tiers. A Jetson Orin Nano Super does perception and planning — stereo depth, visual-inertial SLAM, local reasoning. It is non-real-time by design and allowed to crash. Beneath it an STM32 brainstem holds reflexes, power sequencing and watchdogs, and stays alive when the Jetson does not. Sensing is strictly passive, so several units can share a space without blinding one another.



The complete assembly — mechanical design closed, URDF export next.



Printed hip and limb actuator assembly.

KEY ENGINEERING DECISIONS

Distributed over central control

One node per actuator. Adding a limb is adding nodes, not rewiring the robot, and splitting the chain left/right means one bus fault takes down one side, not the machine.

Passive-only perception

Depth from parallax and scene texture, not a projected dot pattern. Two CatBots can work the same room, and neither announces itself in infrared.

Knees driven by belt

Six knee motors moved up off the distal limb through a 14:1 dual-stage reduction. Limb inertia is what a leg must accelerate every stride, and no controller tuning recovers what the mechanism gives away.

A genuine zero-watt state

The Jetson and every motor controller power down completely while the STM32 draws microamps watching an analogue microphone line. A specific acoustic signature whips the head onto the sound and boots the GPU behind it.

TECHNICAL SPECIFICATIONS

Compute Jetson Orin Nano Super for perception and planning; custom STM32 brainstem for reflexes, power and watchdogs; one STM32G4 FlexNode per actuator.

Actuation 19 DoF — 4 × GIM8108-8 quasi-direct-drive hips, 6 × 5010 360 kV knees on 14:1 S3M belts, 2 × gimbal head motors, 7 × DS3235/DS3230 servos.

Sensing OAK-D Lite passive stereo, distributed VL53L5CX time-of-flight, Knowles MEMS mics into the ultrasonic band, per-node IMU and magnetic encoders.

Power 8S2P Molicel INR21700-P45B — sixteen cells, ~29.6 V nominal, 266 Wh.

Bus CAN-FD at 1 Mbit arbitration / 5 Mbit data, carrying the moteus register protocol.

Structure Forged carbon fibre and titanium as production intent; 3D-printed structure and CNC limb mounts in hand.

Status Design closed, torso assembled; URDF export and node bring-up in progress. Nothing here is validated end-to-end.

FlexNode

DISTRIBUTED CAN-FD COMPUTE NODE

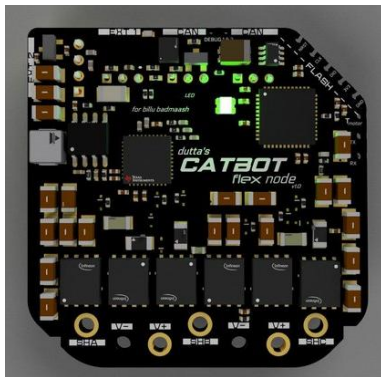
OBJECTIVE

Collapse mobile-robot electronics down to one repeatable part, so that any sensor-fused robot reduces to buying FlexNodes, actuators and sensors, wiring power and CAN, and developing no electronics at all.

SYSTEM OVERVIEW

FlexNode fuses a high-current field-oriented motor controller and a sensor-fusion front end onto one compact PCB hanging off a CAN-FD bus. It began as a fork of the open-source mjbots moteus r4.11 — proven power stage, DRV8353S gate driver and STM32G4 core kept intact — and adds what a distributed node needs and a bare motor controller does not: a 6-axis IMU, optional multizone time-of-flight ranging, a servo/aux port with its own 5 V current sense, an I²C port for load cells or secondary encoders, addressable status RGB, and an onboard 5 V / 3 A supply.

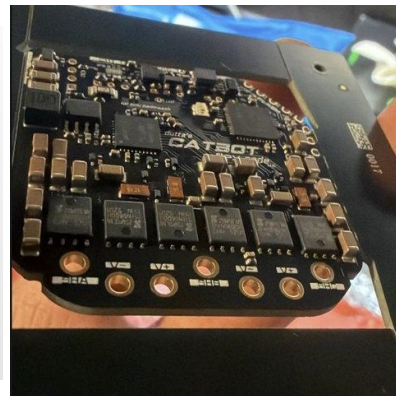
The fleet runs one firmware image. Node identity — CAN ID, encoder sources, attached peripherals — is per-node configuration rather than code, so thirteen joints means thirteen identical boards, two wires between them and a config file each.



FlexNode v1.0 – front side



FlexNode v1.0 – back side I/O



First article — STM32G4, DRV8353S and the six-FET power stage — inspected, not yet powered.

KEY ENGINEERING DECISIONS

One image, N configurations

A reserved register block lets a node advertise the peripherals it actually carries, so the host discovers the fleet rather than being told about it.

Deleted the autodetection

Upstream probes strap pins and the gate driver at boot to identify the board. FlexNode repurposes those pins, so identity is hardcoded and the probe removed outright — ~1.7 KiB of flash returned, and boot provably never drives a repurposed pin.

A two-line fix instead of a respin

The v1.0 copper swaps two motor phases against the reference. Rather than rework every board or edit the control core, the drive-side pin mapping is swapped in the hardware-description layer; the FOC code stays byte-for-byte upstream.

Streaming what will not fit

The time-of-flight sensor needs an ~84 KiB init blob against ~36 KiB of free flash. It is streamed from the host over the CAN diagnostic tunnel at boot rather than stored.

TECHNICAL SPECIFICATIONS

Microcontroller STM32G473CEU6 — Cortex-M4F at 170 MHz, 512 KB flash, UFQFPN-48.

Gate driver & power TI DRV8353S driving six 60 V / 1.6 mΩ N-channel MOSFETs; motor bus to 44 V.

Feedback AS5047P on-axis magnetic encoder over SPI, plus per-phase current sensing.

Onboard sensing LSM6DS3TR-C 6-axis IMU; optional VL53L7CX 8×8 multizone time-of-flight.

Communications CAN-FD via TCAN334G at 1 Mbit / 5 Mbit, daisy-chained, 120 Ω termination on leaf nodes only.

Logic rails LGS5160C synchronous buck to 5 V / 3 A, ME6216A33 LDO to 3.3 V.

Protocol moteus register protocol plus a FlexNode block at 0x080–0x0FF for capability discovery.

Licence Apache-2.0 — all sources at github.com/thedutta/FlexNode.

Status v1.0 assembled and inspected; firmware layer building; bring-up pending.

Sylvara™

ULTRA-LOW-POWER MODULAR PLANT-CARE ECOSYSTEM

OBJECTIVE

Design and deploy a compact, modular plant-care system capable of year-scale autonomous operation, retrofittable into existing gardening and agricultural setups, under extreme power, size and cost constraints.

SYSTEM OVERVIEW

Sylvara is a production-oriented IoT platform aimed at long-term unattended deployment rather than short-lived smart-home demonstrations. It integrates environmental sensing, soil monitoring, energy harvesting and wireless connectivity into a very compact form factor, designed as a modular ecosystem around a head unit currently in beta testing, and prioritising ultra-low power consumption, field reliability and manufacturability.

The company was incorporated in Canada, originally as Smart Plant Inc. and now trading as Sylvara Inc. (BCA Articles of Incorporation, OCN 1001215100, 23 April 2025). It received CAD 10,000 in funding in the summer of 2025, is trademarked and registered, and continues under the support of the Morrissette Institute for Entrepreneurship with active venture interest. The first production batch of 100 units is in manufacture.



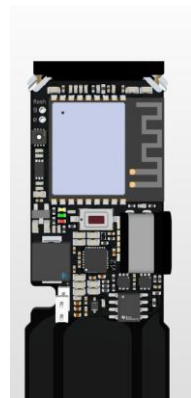
Render in use with optional watering module — solar harvesting, no wiring, no scheduled maintenance.



v1.2 board in hand, beside its CAD model.



V1.1 Render.



V1.2 PCB Top View.

TECHNICAL SPECIFICATIONS

Microcontroller ESP32-C3-WROOM-02 — Wi-Fi, BLE, RISC-V.

Power system TI BQ25798 Li-ion management, solar harvesting, USB-C PD charging.

Battery life Multi-month to roughly a year in zero sunlight; effectively unlimited on twenty minutes of sun a day.

Sensing Temperature, humidity, light, soil moisture and capacitive water-level monitoring.

PCB Approximately 22 × 35 mm, multi-layer integrated design.

Connectivity Wi-Fi and BLE with cloud backend integration.

Status v1.2 released to production; first batch of 100 units in manufacture.

KEY ENGINEERING DECISIONS

Ultra-low-power system design

Firmware and hardware were architected together around aggressive sleep cycles, peripheral power gating and duty-cycled sensing, so that near-perpetual operation follows from roughly twenty minutes of daily sunlight rather than from a larger battery.

High-density integration

Sensing, power management, charging and compute were consolidated onto a single compact PCB through careful IC selection, minimising footprint while preserving reliability and serviceability.

Energy-first architecture

Power budgeting and energy flow were treated as first-class design constraints, shaping component selection, firmware scheduling and communication strategy rather than being optimised afterwards.

Design for production

The system was designed for scalable manufacturability from the outset — optimised part dependencies, waterproof enclosure integration, and a bill of materials that survives contact with a real assembler.

REVISION HISTORY

Three board revisions, and what each one taught. Recording these honestly is the point: the failures are where the engineering actually happened.

v1.0

- Did not take advantage of the ESP32's built-in USB debug functionality, GPIOs had been shared elsewhere.
- Dual-PCB architecture abandoned on cost and assembly complexity.
- Hobby-grade TP4056 charging IC abandoned for thermal, space and telemetry reasons; moved to an industry-grade TI part.
- Passive component footprints left unstandardised, and needed to be.

v1.1

- Charging IC (TI BQ25798) configured incorrectly and manually bypassed in order to test operation.
- GPIO-based capacitive sensing proved non-linear and prone to RTOS timing issues; replaced with a TLC555 timer-based circuit.
- 10 μ F 0402 capacitors proved both a sourcing and a performance problem; moved to a 1206 package for those alone.

v1.2

- EMI modelled incorrectly in MATLAB, but measured within acceptable spec on the bench.
- ESD/TVS diodes missed on the USB data lines.
- USB-C IPX9 gasket did not fit, from misreading the tolerances in the datasheet.

COMPLETED PROJECT

RoVolt

WHEG ROBOT & MOBILE AGRICULTURAL PLATFORM

OBJECTIVE

Build a terrain-adaptive robotic platform capable of reliable locomotion across uneven farmland, then extend it into a smart agricultural system supporting soil interaction, sensing and basic farming operations.

SYSTEM OVERVIEW

RoVolt began as a Class 11 project built around a WHEG — wheel-leg hybrid — locomotion system. The platform was fully CAD-modelled and fabricated from laser-cut aluminium, manually brazed into a rigid but serviceable chassis. Locomotion came from custom 3D-printed WHEGs on a push-rod suspension, driven by DC motors with encoder feedback under a Raspberry Pi.

It demonstrated stable traversal across varied terrain including loose soil. For competitions the system was reconfigured from the original 2×3 WHEG layout into a three-wheel configuration and fitted with a rear-mounted arm carrying soil-moisture sensing, ploughing and seeding attachments. An IMU added directional awareness while the existing encoders were reused for motion feedback, and a companion mobile application enabled real-time monitoring. I led hardware integration and low-level firmware. The platform was shown at multiple expos and competitive hackathons, taking top placements and awards, and was covered in the press.



Laser-cut aluminium, brazed by hand.



Assembled chassis, electronics bay open.



Press coverage, Hackathon '23.

TECHNICAL SPECIFICATIONS

Microprocessor Raspberry Pi 3 on a proto-board.

Control TB6612FNG dual motor drivers with quadrature encoders.

Locomotion Custom 3D-printed WHEGs on a push-rod suspension.

Sensors Wheel encoders, IMU, soil-moisture sensor, arm kinematics.

Fabrication Laser-cut aluminium chassis, brazed by hand; 3D-printed mounts and WHEGs.

KEY ENGINEERING DECISIONS

WHEG-based locomotion

A wheel-leg hybrid balances mechanical simplicity against terrain adaptability far better than a conventional wheeled platform.

Hands-on fabrication

Designing, laser-cutting and brazing the chassis myself kept structural integrity and iteration speed in my own hands rather than a supplier's.

Platform adaptability

Reconfiguring the locomotion layout and adding a modular agricultural arm demonstrated versatility beyond mobility alone.

Sensor reuse and expansion

Existing encoders were reused while inertial sensing was added — better directional stability, no extra complexity.

Mentorship

LEADERSHIP & TECHNICAL DIRECTION

As President of RADAR, one of the most active technical clubs at Manipal Institute of Technology Bengaluru, I mentor multiple student teams, translating ambitious ideas into technically feasible systems against real skill and resource constraints. The role is architectural guidance, design reviews, subsystem decomposition, and helping teams close the gap between concept and execution.

ONGOING MENTORED PROJECTS

TrackMate An autonomous athletic-track pacer focused on consistent velocity control and lap accuracy. In collaboration with Tenvik Sports.

Arachne A spider-robot reconnaissance platform exploring bio-mimetic locomotion and terrain adaptability — a scaled-down mirror of CatBot.

BattleRobot team Competitive combat robotics: high-current power systems, mechanical durability and failure-resilient design.

CO-DESIGNED & COMPLETED

DoorThing — retrofit electronic door unlock

A low-cost, ID-card-based door unlock retrofit built around reliability, power efficiency and ease of installation — embedded control and secure actuation under real institutional constraints.

RC vehicle platform — event robotics kit

An iterable, modular RC platform for a student robotics event, robust enough to be repurposed for the BattleRobot at short notice. Built on Arduino and NRF modules around a compact interconnection PCB, it became the teaching entry point for drivetrain fundamentals and motor control.

CSEA Academic Hub

A student-facing hub for CSE Section A, grouping tools, materials and presentations in one place.



Club orientation day, 2025.



TrackMate on the track.



BattleRobot drivetrain on the bench.

OUTSIDE ENGINEERING

I maintain long-distance endurance training and regular exercise as a way of sustaining discipline, recovery and long-term cognitive performance. I currently run with the track team, and placed third in a recent half marathon.



Conclusion

ENGINEERING TRAJECTORY & INTENT

My work in robotics follows a lineage rather than a collection of independent projects. RoVolt was the origin point: a WHEG-based mobile robot that demonstrated reliable locomotion across uneven terrain and was later adapted into a functional agricultural platform with sensing, actuation and field interaction. Building it end to end — mechanically fabricated, electrically integrated, iteratively modified — established my grounding in how robots physically engage with real environments, and its results across expos and hackathons validated both the platform and a systems-level way of working.

From RoVolt came two deliberate and divergent evolutions among my projects. Sylvara is the agricultural branch, shifting the core challenge from mobility to endurance, integration and scalability inside a funded startup: energy-first thinking, ultra-low-power electronics, dense PCB integration and production-oriented design. CatBot is the autonomy branch, pushing toward agile locomotion, perception-guided decision-making and real-time control under strict size, weight and reliability constraints.

FlexNode is what happened when CatBot's hardest constraint was taken seriously enough to become its own product. Nineteen joints could not each be wired back to one controller, so the electronics were pushed out to the joints and made identical, configurable and additionally open source. It is the piece I would hand to someone else: buy the nodes, wire power and CAN, and skip the year of electronics development that otherwise stands between an idea for a robot and a robot.

Together these reflect a progression in both technical scope and engineering responsibility — from a mobile robotic platform, to a commercial-grade embedded system, to a research-driven autonomous robot, to a reusable piece of open hardware. The through line is a preference for deployable systems, constraint-aware design and holistic integration over isolated technical novelty.

Moving forward, my goal is to keep building autonomous robotic systems that extend human capability in environments where risk, scale or endurance limit direct human presence.

Documentation for every project above, including full hardware and firmware sources, is published at thedutta.github.io.

PROJECT INDEX

CatBot Biomimetic quadruped, 19 DoF, 4.8 kg, CAN-FD spine.

thedutta.github.io/projects/catbot

FlexNode Distributed CAN-FD compute node. Apache-2.0.

github.com/thedutta/FlexNode

Sylvara Ultra-low-power plant care. Incorporated, in production.

sylvara.life

RoVolt WHEG robot and mobile agricultural platform.

thedutta.github.io/projects/rovolt

RADAR TrackMate, Arachne, BattleRobot, DoorThing, RC platform.

thedutta.github.io/projects

Notes Engineering write-ups and process notes.

thedutta.github.io/blog